- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

7 The velocity of a particle moving in a straight line is modelled by $v = 0.6t^2 - 2.1t + 1.5$ where v is the velocity in metres per second and t is the time in seconds.

(a) Determine the times at which the particle is stationary. **[2]**

(b) Find the acceleration of the particle at the first of the times at which it is stationary. **[2]**

(c) Find the distance travelled by the particle between the times at which it is stationary. **[2]**

7(a)

7(b)

7(c)